

Overview

The code in this replication package constructs the analysis files, conducts analyses, and generates the figures and tables for “What Impacts Can We Expect from School Spending Policy? Evidence from Evaluations in the U.S.” (Jackson & Mackevicius, 2023) using Stata and R.

Data Availability and Provenance Statements

This paper does not involve analysis of external data. Author-generated (or webscraped) data are included in these provided files:

- CapitalProjectsYrXYr.xls
- SchoolSpendingPapers.xlsx
- CPIData.dta (from publicly-avaialble FRED database)
- SchoolSpendingPapers_overall.dta (dataset used for the majority of analyses)
 - Note: This is generated in DatasetCreator.do, and can be used to create figures and tables (see table below specifying output .tex and .pdf fragments)
- Additional datasets used for sensitivity and robustness (in /Created Datasets folder)
 - Note these are generated manually from code output:
 - corrbarchart_estimates.xlsx
 - pubbiasbarchart_estimates.xlsx
 - ests_forbayes.xlsx

Statement about Rights

I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

All data **are** publicly available.

Details on each Data Source

Data.Name	Data.Files	Location	Provid ed	Citation
Estimates Captured for “What Impacts Can We Expect from School Spending Policy? Evidence from Evaluations in the U.S.” (Jackson & Mackevicius, 2023)	CapitalProjectsYrXYr.xls; SchoolSpendingPapers.xlsx	InputData/	TRUE	Jackson & Mackevicius (2023)
CPI Data	CPIData.dta	InputData/	TRUE	U.S. Bureau of Labor Statistics

Computational requirements

Software Requirements

- Stata (code was last run with version 16.1)
 - estout
 - robumeta
- R (code was last run with version 4.0.4)
 - foreign
 - decon
 - meta
 - haven
 - deconvolveR
 - ggplot2
 - REBayes
 - metaplust
 - readr
 - xtable

Controlled Randomness

- Set seed immediately before bootstrap tau

Memory and Runtime Requirements

It should take no more than 10 minutes to reproduce analyses on a standard (2022) desktop machine.

Summary

Approximate time needed to reproduce the analyses on a standard (2023) desktop machine: 10-60 minutes.

Details

The code was last run on a laptop with MacOS version 13.4.1. Computation took less than 60 minutes.

Description of programs/code

- capital_input.do:
 - Reads in CapitalProjectsYrXYr.xls, constructs estimates for year 0-X for each capital spending paper. These are the estimates for capital papers, equivalent to the estimates for non-capital papers (that are captured from each paper and described in Table A.1)
 - Uses CPIData (downloaded from Federal Reserve) to CPI-adjust \$ amounts.
- overall_input.do:
 - Reads in SchoolSpendingPapers.xlsx, constructs estimates for each study-outcome and generates datasets that will be read in for analyses.
 - This involves calling DatasetCreator.do to create datasets for the overall sample (SchoolSpendingPapers_overall), as well as set of subsamples and assumption bounds for sensitivity and

robustness, as well as .csv files for one web-application-based test of publication bias.

■ Output datasets are written into folder “/Created Datasets”

- MainAnalysis.do:
 - Reads in SchoolSpendingPapers_overall, generates main table (Table 3) and figure (Figure 6).
- FiguresTables.do:
 - Reads in SchoolSpendingPapers_overall (or _<sensitivity/robustness> datasets), generates most figures and tables.
- bayes.do:
 - Generates bayes estimates
 - Creates tables comparing bayes to overall estimate
- metaplus.R:
 - generates estimates for Table A.5
 - note: reads in .csv generated in SchoolSpendingAnalysis.do
- metawork.R:
 - run Shapiro-Wilk test and create Q-Q plots
- small_deconv.code.R:
 - generates deconvolve estimates
- DeconvolvePolicyFig.do
 - generates deconvolve policy figures
- metaanalysis_table.do:
 - creates AnalysisPapersTable
- prosesteps_table.do:
 - creates PerPaperSteps table
- LInonLI_table.do:
 - creates LInonLICompare table
- capital_justificationtable.do:
 - creates CapitalDepJustification table
- SimpleDeconvolve.R:
 - generates deconvolved estimates
- show_normal_deconv.do:
 - generates normality plot (w/ CI) of deconvolve estimates
 - Note: run small_deconve.R first

Instructions to Replicators

1. Each Stata do-file begins with a place to adjust the local filepath, expecting these subfolders exist at that filepath:
 - a. BS Data (this will be populated)
 - b. Created Datasets (this will be populated, contains the last version of these datasets we used for analyses)
 - c. Figures and Tables (this will be populated)
 - d. Input Data (populated with provided data)

- e. R Code
- f. Stata Code
- 2. Each short R script includes filepaths to update within (sometimes in multiple places)
- 3. You may replicate from “raw” excel data (Jackson & Mackevicius, 2023) (step 1 below), or use the provided SchoolSpendingPapers_overall.dta and replicate the analyses, figures, and tables (step 2 below). Tables are constructed into .tex fragments and figures into .pdf files. Exceptions include Figures A.1 A.17, A.18 which are captured from screenshots and Table A.5 which is constructed from the R output in metaplus.R.

Details

1. If replicating from “raw” data (excel sheet with author-captured estimates for each paper), take these steps to create datasets for analysis:
 - a. Run capital_input.do, which generates amortized estimates for capital spending papers—these are already in SchoolSpendingPapers.xlsx
 - b. Run overall_input.do, which calls DatasetCreator and generates (into folder /Created Datasets) all datasets for analyses
 - c. Proceed to next step
2. If replicating from generated datasets (skipping step 2), take these steps to replicate analyses, figures, and tables:
 - a. Run config.do to install necessary packages
 - b. Run MainAnalysis for the main table (Table 3) and figure (Figure 6)
 - c. Run FiguresTables.do to generate most figures and tables (See table below)
 - d. Run additional .do and .R code for figures and tables not included in FiguresTables (see table below)
 - i. These include analyses reliant on R commands
 - ii. These include .tex tables generated with lengthy Stata code that would be cumbersome if included in FiguresTables.do

List of tables and programs

The provided code reproduces:

- All numbers provided in text in the main paper, and
- All tables and figures in the main paper

Figure/Table	Do-file	Line number	Output File (Figures are .pdf files and tables are .tex fragments)
Figure 1	FiguresTables.do	1838-1907	combined_capovertime
Figure 2	FiguresTables.do	1573-1766	multper_testscore_forest //

			multper_nontest_forest
Figure 3	FiguresTables.do	1394-1566	linearinspend_combined ONEPER // returnsbybaselineONEPE R
Figure 4	FiguresTables.do	1910-1944	trimcombined_multper
Figure 5	FiguresTables.do	77-99	pubbiasbar
Figure 6	MainAnalysis.do	18-95	policyprob_testscore // policyprob_nontestscore
Figure 7	DeconvolvePolicyFig.do	1-119	test_deconvolve // edattain_deconvolve
Table 1	FiguresTables.do	121-213	equalweight_regs
Table 2	metaanalysis_table.do	1-96	AnalysisPapersTable
Table 3	MainAnalysis.do	98-390	PolicyProbRegs
Table 4	FiguresTables.do	989-1073	Fstat20_regs
Figure A.2	FiguresTables.do	1793	dateHist
Table A.1	prosesteps_table.do	1	PerPaperSteps
Table A.2	LlnonLI_table.do	1	LlnonLICompare
Figure A.3	FiguresTables.do	1573	overall_testscore_forest // overall_nontestscore_for est
Figure A.4	FiguresTables.do	1771	bstau_comb
Figure A.5	FiguresTables.do	1394	linearinspend_combined MULTPER
Figure A.6	FiguresTables.do	1330	returnsbybaselineMULTP ER
Figure A.7	FiguresTables.do	1948	returnsbyspendchange
Figure A.8	FiguresTables.do	1573	noncapital_testscore_for est
Figure A.9	FiguresTables.do	1573	capital_testscore_forest
Figure A.10	FiguresTables.do	1573	fstat_test_forest
Figure A.11	FiguresTables.do	1573	fstat_edattain_forest

Table A.3	FiguresTables.do	652	dimensions_metaregs
Figure A.12	FiguresTables.do	393	AttainmentOverTimeEst
Table A.4	capital_justificationtable.do	1	CapitalDepJustification
Figure A.13	FiguresTables.do	1822	depexemplar
Figure A.14	metawork.R	1-91	normality_scores_unweighted // normality_scores_weighted
Figure A.15	metawork.R	1-91	normality_edattain_unweighted // normality_edattain_weighted
Figure A.16	DeconvolvePolicyFig.do	1	test_deconvolve // edattain_deconvolve
Figure A.17	show_normal_deconv.do	1	normalityplotCI_scores // normalityplotCI_edattain
Table A.5	(manually constructed from metaplus.R results)	19; 27	metaplustable
Figure A.18	FiguresTables.do	216	sampleeffect
Table A.6	FiguresTables.do	235	sampleeffect_regslnSE
Figure A.19	FiguresTables.do	49	corr_sense_bar
Table A.7	FiguresTables.do	1256	noClustSDIV_sensregs
Table A.8	FiguresTables.do	1180	byCorrSens
Table A.9	FiguresTables.do	892	Titlel_Llregs
Table A.10	FiguresTables.do	1077	withinAcrossSens
Table A.11	FiguresTables.do	793	capdep_sens_regs
Table A.12	FiguresTables.do	696	linearinspend_regs
Table A.13	FiguresTables.do	332	PowerPolicyCatRegs
Figure A.20	FiguresTables.do	77	pubbiasbar
Table A.14	FiguresTables.do Note: manually update trim&fill and Andrews&Kasy	550	PubBiasregs

Figure A.21	FiguresTables.do	2006	trimcombined
Figure A.22	FiguresTables.do	1910	trimcombined_multper
Table A.15	FiguresTables.do	505	PubTypeRegsFINA
Table A.16	FiguresTables.do	460	tstat_regs
Figure A.23	FiguresTables.do	440	t_hist_both
Figure A.24	Captured from: https://maxkasy.github.io/home/metastudy/	n/a	KasyTest
Figure A.25	Captured from: https://maxkasy.github.io/home/metastudy/	n/a	KasyNonTest
Table A.17	bayes.do	1	bayescomp
Table A.18	bayes.do	1	bayesbyfirststage

References

U.S. Bureau of Labor Statistics, Consumer Price Index for All Urban Consumers: All Items in U.S. City Average [CPIAUCSL], retrieved from FRED, Federal Reserve Bank of St. Louis;
<https://fred.stlouisfed.org/series/CPIAUCSL>.

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openicpsr.org/openicpsr/project/182042